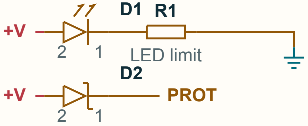
The whole circuit.

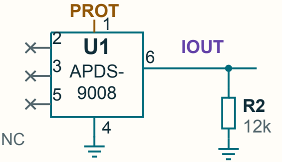
Same label = same wire, even between separate boxes.

1 CONNECT + DECOUPLE

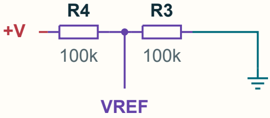
2 POWER + LED



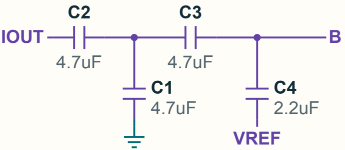
3 SENSOR + CURRENT LOAD



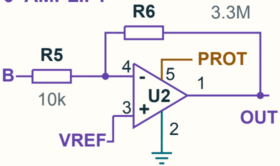
4 MAKE A MIDPOINT



5 FILTER



6 AMPLIFY



GND = 0V PROT = supply after D2 NC = intentionally unconnected

You can read this.

Start with J1. Then follow the matching labels.

TL;DR

J1 brings in power and carries OUT back.

U1 + R2 feed the filter. U2 amplifies the result.

1 **CONNECT + DECOUPLE**
J1 is the cable connector. C5 helps steady the incoming supply. Pins: 1 GND, 2 +V, 3 OUT.

2 **POWER + LED**
D1 is the LED; R1 limits its current. D2 feeds the protected supply, PROT, to U1 and U2.

3 **SENSOR + CURRENT LOAD**
U1 is the sensor chip. R2 turns its output current into a voltage at the IOUT connection.

4 **MAKE A MIDPOINT**
R4 and R3 divide +V in half to make VREF. This reference gives the signal a midpoint.

5 **FILTER**
C2 and C3 are in the signal path. C1 and C4 connect it to GND and VREF, shaping the response.

6 **AMPLIFY**
U2 (MCP6001) amplifies. R6 feeds OUT back to its - input. R5 / R6 set gain; + uses VREF.

THE SYMBOL KEY

R resistor **C capacitor** **D diode / LED** **U chip**

Dot = wires join. Small pin numbers match the chip or connector.

Match R1, C2, U2... to the PCB + BOM on card 01.